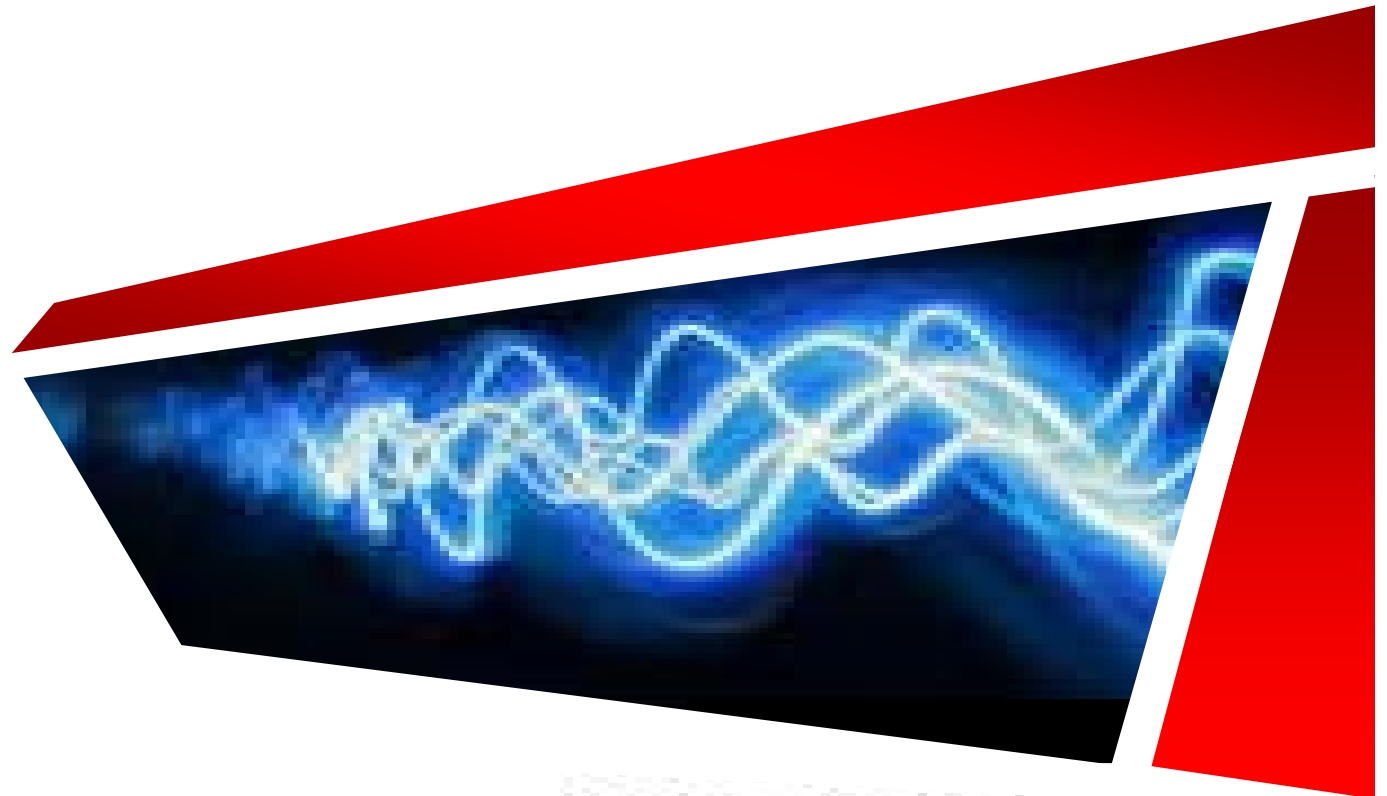


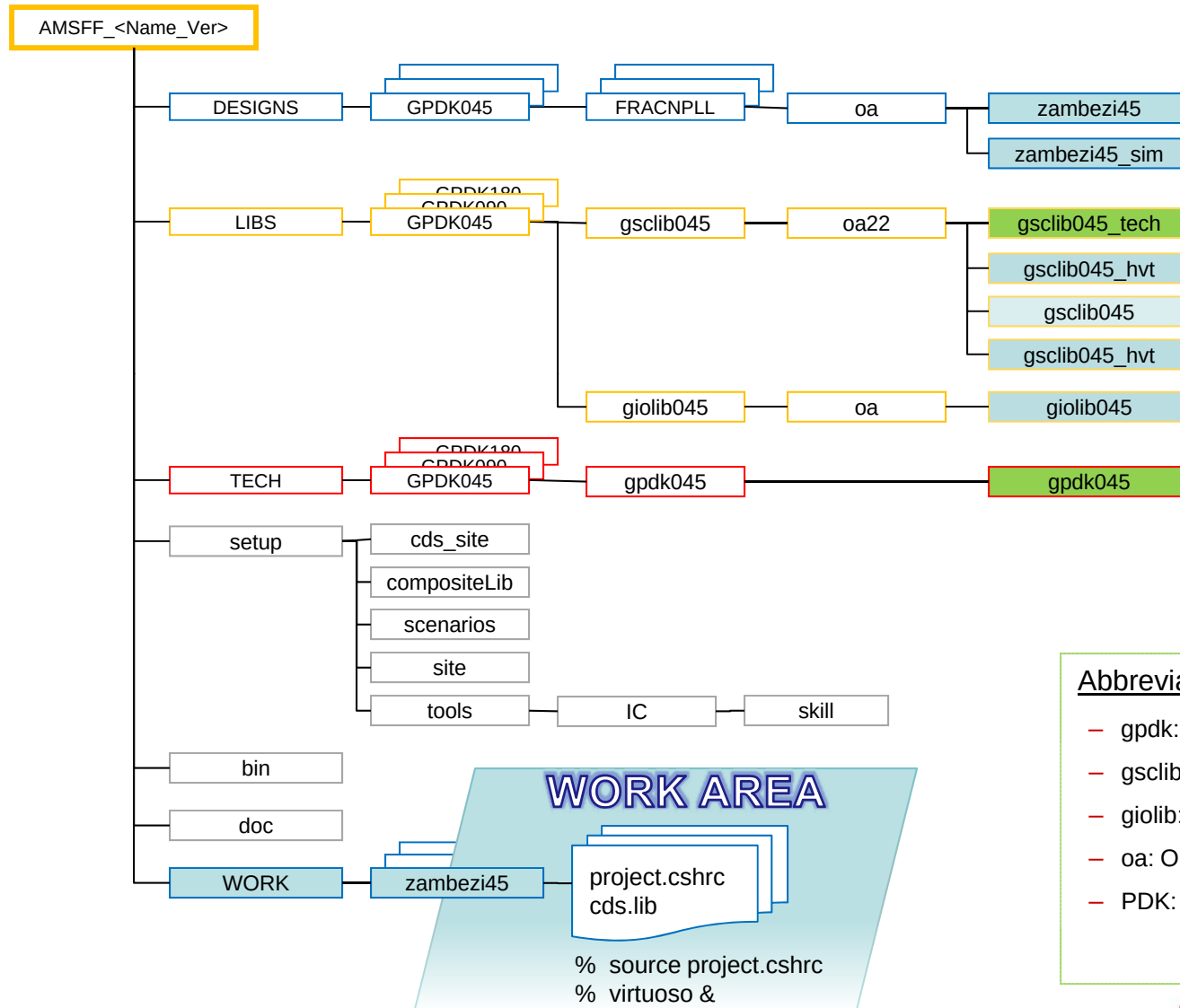
AMSFF Database Diagrams

Revision: v5 (2013-April)



AMSFF DB: Unix/Linux Data Structure

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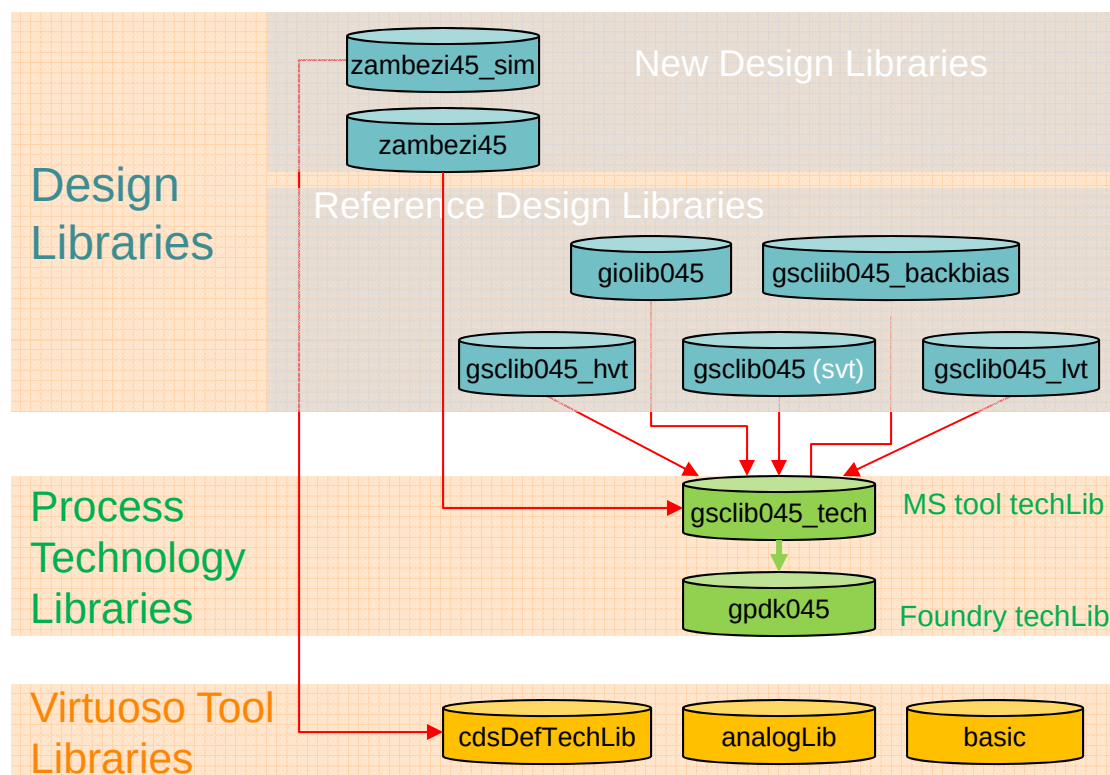


Abbreviation Used:

- gpdk: Generic PDK
- gsclib: Generic Standard Cell Library
- giolib: Generic IO Library
- oa: Open Access (database type)
- PDK: Process Design Kit

AMSFF DB: Virtuoso Library Binding

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Legends:

- Red arrow: Technology library **attaching**
- Green arrow: Incremental technology **referencing**

Info: For detail of two library binding types (**Attach** and **Reference**), see FAQs slide

Notes:

- MS tool techLib contents:
 - refTechLibs() for ITDB
 - Custom via defs (P&R)
 - Constraint groups for P&R
 - Site definitions
 - Layout-XL CG
 - Design project CG
- Foundry techLib contents:
 - Layer and display definitions
 - Basic process constraints
 - Standard via definitions
 - Device, path, and guarding definitions
- Abbreviation Used:
 - gpdk: Generic PDK
 - gsclib: Generic Std. Cell Lib
 - giolib: Generic IO Library
 - hvt: High Vt, lvt: Low Vt

AMSFF DB In Virtuoso Library Manager

The screenshot displays the Cadence Virtuoso Library Manager window. The title bar reads "Library Manager: Directory ...A/WA_rayv/AMSFF/WORK/zambezi45". The menu bar includes "File", "Edit", "View", "Design Manager", "DB Linter", and "Help". The Cadence logo is in the top right corner.

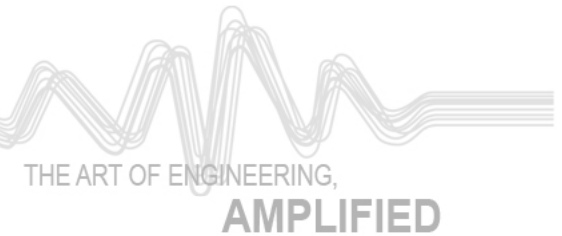
Below the menu bar, there are two checkboxes: ☒ Show Categories and ☐ Show Files.

The interface is divided into four main panes:

- Library:** Shows a hierarchical tree of libraries. The "zambezi45" library is selected, showing sub-libraries like "Cadence_IC", "US_8ths", "ahdLib", "analogLib", "basic", "cdsDefTechLib", "functional", "pllLib", "rfLib", "sample", "FracN_PLL_45", "FracN_PLL_90", "PDKg45n_ITDB", and "PDKg90n_ITDB".
- Category:** Shows a list of categories under "Chip_LowPower", including "Everything", "Uncategorized", "Chip_LowPower", "CustomDigitalBlocks", "TopSubBlocks", "atb", "caps", "cp", "div", "lf_lpf", "ls", "pfd", "pll_dig", "reg", "sheets", and "vco".
- Cell:** Shows a list of cells under "LP_pll_chip", including "LP_pll", "LP_pll_chip", "LP_pll_dig", "LP_pll_dig_combo", "LP_pll_dig_wSPI", "LP_pll_ls_dvdd2avdd", "LP_pll_ls_dvdd2avddh", "clk_gate", "power_manager", "spi", and "test_control".
- View:** Shows a table of views for the selected cell. The table has columns for "View", "Lock", and "Size".

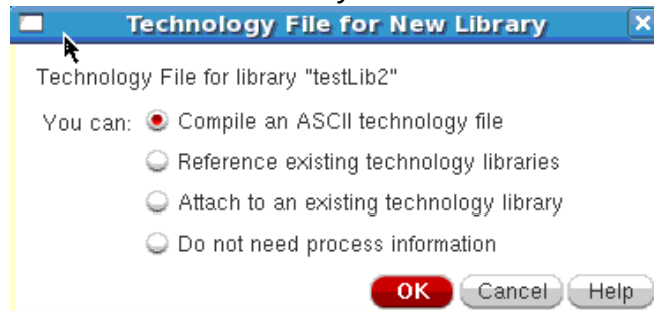
View	Lock	Size
Info_pin_package		4k
constraint		32
layout		535k
layout.routed		261k
layout_fills		65.7M
layout_metalFilled		18k
layout_place		528k
layout_place1		450k
layout_place1a		481k
layout_place2		527k
layout_place4c2		446k
layout_routed		561k
layout_routed_pwr		337k
layout_w_ioring		261k
physConfig		243
schematic		200k
schematic_w_ioring		177k
symbol		36k

AMSFF DB: FAQs



Q1. What is the Virtuoso library *attaching* and *referencing*?

The Virtuoso tool provides these 4 options to users when create a new library:



EDI equivalency:

```
% createOaLib <libName> -referenceTech ...  
% createOaLib <libName> -attachTech ...
```

The '*Reference*' means to compiling this new library into an incremental technology database (ITDB) reference. The lowest one (or last list elements) is the base; and the upper one is the derived technology. This new created library is a new technology with *tech.db* binary file.

The '*Attach*' means to bind the library to an existing technology library; and the *tech.db* file is not created.

The '*Do not need process information*' means to attaching this new library to tool technology library: *cdsDefTechLib*.

Q2. When I should use the '*Do not need process information*' option?

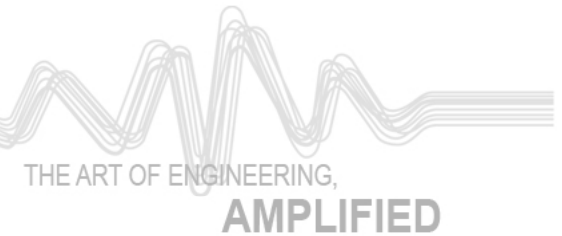
This option is suitable for library without physical implementation (or layout).

Q3. How I change the existing design library on *attach* and *reference*?

For *attach*: Use the *Technology Tool Box* utility (CIW menu: Tools -> Technology Tool Box)

For *Reference*: Create an ASCII techFile with reference library/ies as list element/s in *refTechLibs()* section. Then, load the techFile via *Load* button in the *Technology Tool Box* utility.

AMSFF DB: FAQs



Q4. What is a recommended method for metal option?

- (a) Copy the *gsclib045_tech* to either different name or different path
- (b) Update entry in LDRS and LSRS with the needed metals and vias in techFile of *gsclib045_tech*
- (c) Update *cds.lib* file entry to point to the new library with metal option

For example (cds.lib entry):

```
# define gsclib045_tech <path>/LIBS/GPDK045/gsclib045/oa22/gsclib045_tech
define gsclib045_tech <path>/LIBS/GPDK045/gsclib045/oa22/gsclib045_tech6M1P
# define gsclib045_tech <path>/LIBS/GPDK045/gsclib045/oa22_6M1P/gsclib045_tech
```

Note: There is no change on library attaching and referencing on design libraries.

Q5. What technology library that my analog design library should bind to (*gpdk045* or *gsclib045_tech*)?

Case of no custom routing and no metal restriction, attach to *gpdk045*

Case of either use custom routing (in VLE-XL/GXL) or having metal option, attach to *gsclib045_tech*



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